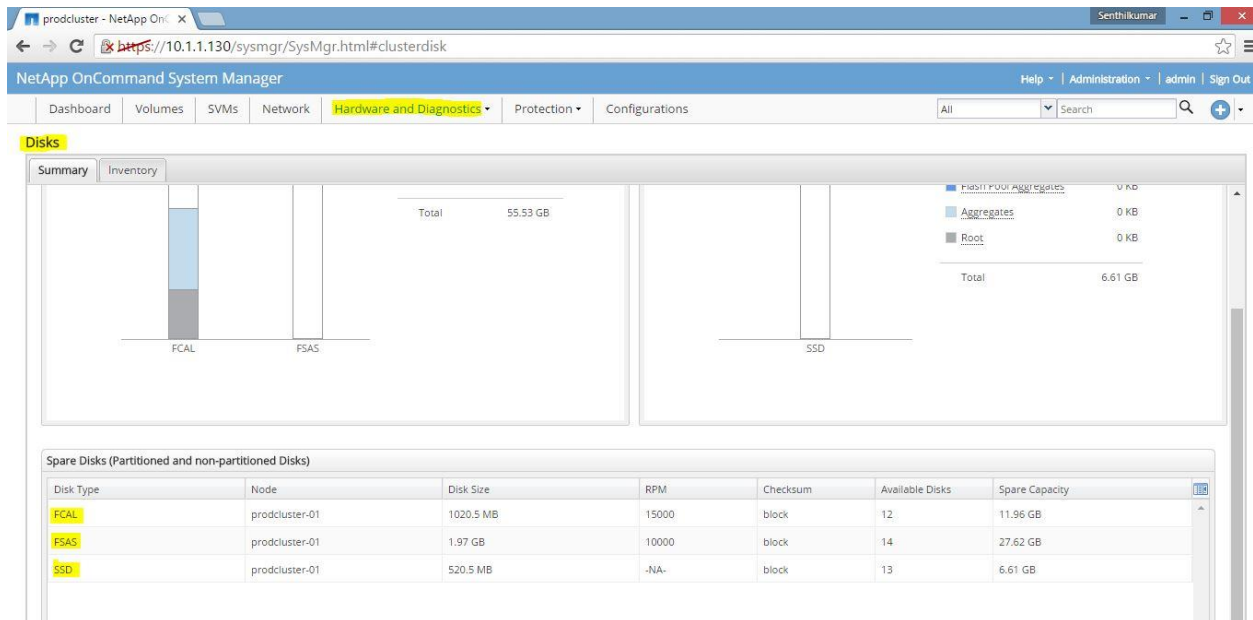


## Disk Management:

### Listing the disks inventory using System Manager



### Listing the SSD disk type alone.

```
prodcluster::>
prodcluster::> storage disk show -type SSD
```

| Disk        | Usable Size | Shelf | Bay | Disk Type | Container Type | Container Name | Owner        |
|-------------|-------------|-------|-----|-----------|----------------|----------------|--------------|
| NET-1.1 01  | 520.5MB     | -     | 24  | SSD       | spare          | Pool0          | prodcluster- |
| NET-1.2 01  | 520.5MB     | -     | 25  | SSD       | spare          | Pool0          | prodcluster- |
| NET-1.3 01  | 520.5MB     | -     | 26  | SSD       | spare          | Pool0          | prodcluster- |
| NET-1.4 01  | 520.5MB     | -     | 27  | SSD       | spare          | Pool0          | prodcluster- |
| NET-1.5 01  | 520.5MB     | -     | 28  | SSD       | spare          | Pool0          | prodcluster- |
| NET-1.6 01  | 520.5MB     | -     | 29  | SSD       | spare          | Pool0          | prodcluster- |
| NET-1.7 01  | 520.5MB     | -     | 32  | SSD       | spare          | Pool0          | prodcluster- |
| NET-1.51 01 | 520.5MB     | -     | 17  | SSD       | spare          | Pool0          | prodcluster- |
| NET-1.52    | 520.5MB     | -     | 18  | SSD       | spare          | Pool0          | prodcluster- |

## Listing the FCAL typeDisks.

```
prodcluster::>
prodcluster::> storage disk show -type FCAL
```

| Disk       | Usable<br>Size | Shelf | Bay | Disk<br>Type | Container<br>Type | Container<br>Name | Owner        |
|------------|----------------|-------|-----|--------------|-------------------|-------------------|--------------|
| NET-1.801  | 1020MB         | -     | 16  | FCAL         | aggregate         | aggr1             | prodcluster- |
| NET-1.901  | 1020MB         | -     | 17  | FCAL         | aggregate         | aggr1             | prodcluster- |
| NET-1.1001 | 1020MB         | -     | 18  | FCAL         | aggregate         | aggr1             | prodcluster- |
| NET-1.1101 | 1020MB         | -     | 16  | FCAL         | aggregate         | aggr0             | prodcluster- |
| NET-1.1201 | 1020MB         | -     | 17  | FCAL         | aggregate         | aggr0             | prodcluster- |
| NET-1.1301 | 1020MB         | -     | 18  | FCAL         | aggregate         | aggr0             | prodcluster- |
| NET-1.1401 | 1020MB         | -     | 19  | FCAL         | aggregate         | aggr1             | prodcluster- |

## Listing the unassigned disks.

```
prodcluster::> storage disk show -container-type unassigned
```

| Disk     | Usable<br>Size | Shelf | Bay | Disk<br>Type | Container<br>Type | Container<br>Name | Owner |
|----------|----------------|-------|-----|--------------|-------------------|-------------------|-------|
| NET-1.45 | -              | -     | 26  | ATA          | unassigned        | -                 | -     |
| NET-1.46 | -              | -     | 27  | ATA          | unassigned        | -                 | -     |
| NET-1.47 | -              | -     | 28  | ATA          | unassigned        | -                 | -     |

3 entries were displayed.

```
prodcluster::>
prodcluster::>
```

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https://10.1.1.130/sysmgr/SysMgr.html#clusterdisk

NetApp OnCommand System Manager Help Administration admin Sign Out

Dashboard Volumes SVMs Network Hardware and Diagnostics Protection Configurations All Search

Disks

Summary Inventory

Assign Zero Spares Refresh

| Name     | Container Type | Home Owner     | Current Owner  | Type | RPM  | Effective Size | Physical Space |
|----------|----------------|----------------|----------------|------|------|----------------|----------------|
| NET-1.43 | Spare          | prodcluster-01 | prodcluster-01 | ATA  | 7200 | 1.97 GB        | 1.98 GB        |
| NET-1.44 | Spare          | prodcluster-01 | prodcluster-01 | ATA  | 7200 | 1.97 GB        | 1.98 GB        |
| NET-1.45 | Unassigned     | -NA-           | -NA-           | ATA  | 7200 |                | 1.98 GB        |
| NET-1.46 | Unassigned     | -NA-           | -NA-           | ATA  | 7200 |                | 1.98 GB        |
| NET-1.47 | Unassigned     | -NA-           | -NA-           | ATA  | 7200 |                | 1.98 GB        |
| NET-1.48 | Spare          | prodcluster-01 | prodcluster-01 | ATA  | 7200 | 1.97 GB        | 1.98 GB        |
| NET-1.49 | Spare          | prodcluster-01 | prodcluster-01 | ATA  | 7200 | 1.97 GB        | 1.98 GB        |
| NET-1.51 | Spare          | prodcluster-01 | prodcluster-01 | SSD  | -NA- | 520.5 MB       | 527.56 MB      |
| NET-1.52 | Spare          | prodcluster-01 | prodcluster-01 | SSD  | -NA- | 520.5 MB       | 527.56 MB      |

Assigning the disks to the node.

```

10.1.1.130 - PuTTY
prodcluster::> storage disk show -container-type unassigned
      Usable      Disk      Container      Container
Disk      Size Shelf Bay Type      Type      Name      Owner
-----
NET-1.45      -      -   26 ATA      unassigned      -      -
NET-1.46      -      -   27 ATA      unassigned      -      -
NET-1.47      -      -   28 ATA      unassigned      -      -
3 entries were displayed.

prodcluster::>
prodcluster::>
prodcluster::>
prodcluster::>
prodcluster::>
prodcluster::> storage disk assign -disk NET-1.45 -owner prodcluster-01

prodcluster::> storage disk show -container-type unassigned
      Usable      Disk      Container      Container
Disk      Size Shelf Bay Type      Type      Name      Owner
-----
NET-1.46      -      -   27 ATA      unassigned      -      -
NET-1.47      -      -   28 ATA      unassigned      -      -
2 entries were displayed.

```

## Assigning disks using System Manager.

NetApp OnCommand System Manager

Disks

Summary Inventory

Assign Zero Spares Refresh

Assign Disks

Disks selected: 1

A disk is assigned to a node in an HA pair to which it is physically connected. The disk becomes spare after it is assigned to the node.

Assign To:

Assign Cancel

| Name     | Container Type | Home Owner     | Current Owner  | Type | RPM  | Effective Size | Physical Space |
|----------|----------------|----------------|----------------|------|------|----------------|----------------|
| NET-1.43 | Spare          | prodcluster-01 | prodcluster-01 | ATA  | 7200 | 1.97 GB        | 1.98 GB        |
| NET-1.44 | Spare          | prodcluster-01 |                |      |      | 1.97 GB        | 1.98 GB        |
| NET-1.45 | Spare          | prodcluster-01 |                |      |      | 1.97 GB        | 1.98 GB        |
| NET-1.46 | Unassigned     | -NA-           |                |      |      |                | 1.98 GB        |
| NET-1.47 | Unassigned     | -NA-           |                |      |      |                | 1.98 GB        |
| NET-1.48 | Spare          | prodcluster-01 |                |      |      | 1.97 GB        | 1.98 GB        |
| NET-1.49 | Spare          | prodcluster-01 |                |      |      | 1.97 GB        | 1.98 GB        |
| NET-1.51 | Spare          | prodcluster-01 |                |      |      | 520.5 MB       | 527.56 MB      |
| NET-1.52 | Spare          | prodcluster-01 |                |      |      | 520.5 MB       | 527.56 MB      |

Details

Aggregate: -NA-  
Vendor ID: NETAPP  
Zeroing (%): -NA-  
Serial No: 07640410  
Broken Details: unassigned

## Unassigning the disks.

```
prodcluster::>
prodcluster::> storage disk removeowner -disk NET-1.45
prodcluster::> storage disk removeowner -disk NET-1.46
prodcluster::> storage disk show -container-type unassigned
      Usable      Disk      Container      Container
Disk      Size Shelf Bay Type      Type      Name      Owner
-----
NET-1.45      -      -   26 ATA      unassigned      -      -
NET-1.46      -      -   27 ATA      unassigned      -      -
NET-1.47      -      -   28 ATA      unassigned      -      -
3 entries were displayed.
prodcluster::>
```

## Listing the aggregate disks.

```
prodcluster::>
prodcluster::> storage disk show -aggregate aggr1
```

| Disk           | Usable<br>Size | Shelf | Bay | Disk<br>Type | Container<br>Type | Container<br>Name | Owner        |
|----------------|----------------|-------|-----|--------------|-------------------|-------------------|--------------|
| NET-1.8<br>01  | 1020MB         | -     | 16  | FCAL         | aggregate         | aggr1             | prodcluster- |
| NET-1.9<br>01  | 1020MB         | -     | 17  | FCAL         | aggregate         | aggr1             | prodcluster- |
| NET-1.10<br>01 | 1020MB         | -     | 18  | FCAL         | aggregate         | aggr1             | prodcluster- |
| NET-1.14<br>01 | 1020MB         | -     | 19  | FCAL         | aggregate         | aggr1             | prodcluster- |
| NET-1.15<br>01 | 1020MB         | -     | 20  | FCAL         | aggregate         | aggr1             | prodcluster- |

5 entries were displayed.

```
prodcluster::>
```

## Listing the spare disks.

```
prodcluster::> storage disk show -container-type spare
```

| Disk          | Usable<br>Size | Shelf | Bay | Disk<br>Type | Container<br>Type | Container<br>Name | Owner        |
|---------------|----------------|-------|-----|--------------|-------------------|-------------------|--------------|
| NET-1.1<br>01 | 520.5MB        | -     | 24  | SSD          | spare             | Pool0             | prodcluster- |
| NET-1.2<br>01 | 520.5MB        | -     | 25  | SSD          | spare             | Pool0             | prodcluster- |
| NET-1.3<br>01 | 520.5MB        | -     | 26  | SSD          | spare             | Pool0             | prodcluster- |
| NET-1.4<br>01 | 520.5MB        | -     | 27  | SSD          | spare             | Pool0             | prodcluster- |
| NET-1.5<br>01 | 520.5MB        | -     | 28  | SSD          | spare             | Pool0             | prodcluster- |
| NET-1.6<br>01 | 520.5MB        | -     | 29  | SSD          | spare             | Pool0             | prodcluster- |
| NET-1.7       | 520.5MB        | -     | 32  | SSD          | spare             | Pool0             | prodcluster- |

## Listing the disk status.

```
10.1.1.130 - PuTTY
prodcluster:> storage aggregate show-status -aggregate aggr1

Owner Node: prodcluster-01
Aggregate: aggr1 (online, raid4) (block checksums)
Plex: /aggr1/plex0 (online, normal, active, pool0)
RAID Group /aggr1/plex0/rg0 (normal, block checksums)

      Position Disk                Pool Type      RPM      Usable Physical
atus -----
-----
      parity  NET-1.8                0    FCAL    15000    1020MB    1.00GB (n
ormal)
      data    NET-1.14                0    FCAL    15000    1020MB    1.00GB (n
ormal)
      data    NET-1.9                  0    FCAL    15000    1020MB    1.00GB (n
ormal)
      data    NET-1.15                0    FCAL    15000    1020MB    1.00GB (n
ormal)
      data    NET-1.10                0    FCAL    15000    1020MB    1.00GB (n
ormal)
5 entries were displayed.
prodcluster:>
```

## Disk replacement.

```
10.1.1.130 - PuTTY
prodcluster:> storage disk replace -disk NET-1.14 -action start -replacement NET-1.30

prodcluster:> storage aggregate show-status -aggregate aggr1

Owner Node: prodcluster-01
Aggregate: aggr1 (online, raid4) (block checksums)
Plex: /aggr1/plex0 (online, normal, active, pool0)
RAID Group /aggr1/plex0/rg0 (normal, block checksums)

      Position Disk                Pool Type      RPM      Usable Physical
atus -----
-----
      parity  NET-1.8                0    FCAL    15000    1020MB    1.00GB (n
ormal)
      data    NET-1.14                0    FCAL    15000    1020MB    1.00GB (r
eplacing, copy in progress)
      copy    NET-1.30                0    FCAL    15000    1020MB    1.00GB (c
opy 1% completed)
      data    NET-1.9                  0    FCAL    15000    1020MB    1.00GB (n
ormal)
      data    NET-1.15                0    FCAL    15000    1020MB    1.00GB (n
```

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← → C <https://10.1.1.130/sysmgr/SysMgr.html#clusteraggregate>

NetApp OnCommand System Manager Help Administration admin Sign Out

Dashboard Volumes SVMs Network Hardware and Diagnostics Protection Configurations All Search

**Aggregates**

Create Edit Add Capacity Add Cache Mirror Delete Status Volume Move Refresh

| Name      | Node           | Type     | Used (%) | Available Space | Used Space | Total Space | Volume Count | Disk Count | Flash Pool |
|-----------|----------------|----------|----------|-----------------|------------|-------------|--------------|------------|------------|
| aggr0     | prodcluster-01 | Standard | 54       | 1.54 GB         | 1.8 GB     | 3.34 GB     | 1            | 6          | -NA-       |
| aggr1     | prodcluster-01 | Standard | 1        | 3.5 GB          | 20.46 MB   | 3.52 GB     | 1            | 5          | -NA-       |
| depi2aggr | prodcluster-01 | Standard | 1        | 3.5 GB          | 20.45 MB   | 3.52 GB     | 1            | 5          | -NA-       |

| Name     | Disk Type | Physical Size | Usable Size | Disk Status            | Plex Name | Plex Status | Raid Group | Raid type | Storage Pool |
|----------|-----------|---------------|-------------|------------------------|-----------|-------------|------------|-----------|--------------|
| NET-1.8  | FCAL      | 1 GB          | 1020.5 MB   | Normal                 | plex0     | Online      | rg0        | RAID4     | -NA-         |
| NET-1.14 | FCAL      | 1 GB          | 1020.5 MB   | Replacing Copy in p... | plex0     | Online      | rg0        | RAID4     | -NA-         |
| NET-1.30 | FCAL      | 1 GB          | 1020.5 MB   | Copy 61% completed     | plex0     | Online      | rg0        | RAID4     | -NA-         |
| NET-1.9  | FCAL      | 1 GB          | 1020.5 MB   | Normal                 | plex0     | Online      | rg0        | RAID4     | -NA-         |

```

10.1.1.130 - PuTTY
prodcluster::> storage aggregate show-status -aggregate aggr1

Owner Node: prodcluster-01
Aggregate: aggr1 (online, raid4) (block checksums)
Plex: /aggr1/plex0 (online, normal, active, pool0)
RAID Group /aggr1/plex0/rg0 (normal, block checksums)

      Position Disk              Pool Type    RPM    Usable Physical
atus -----
-----
      parity    NET-1.8              0    FCAL    15000    1020MB    1.00GB (n
ormal)
      data      NET-1.30              0    FCAL    15000    1020MB    1.00GB (n
ormal)
      data      NET-1.9              0    FCAL    15000    1020MB    1.00GB (n
ormal)
      data      NET-1.15             0    FCAL    15000    1020MB    1.00GB (n
ormal)
      data      NET-1.10              0    FCAL    15000    1020MB    1.00GB (n
ormal)
5 entries were displayed.

prodcluster::>

```

```
10.1.1.130 - PuTTY
prodcluster::> storage disk show -disk NET-1.30
Disk: NET-1.30
Container Type: aggregate
Owner/Home: prodcluster-01 / prodcluster-01
DR Home: -
Stack ID/Shelf/Bay: - / - / 26
LUN: 0
Array: NETAPP_VD_1
Vendor: NETAPP
Model: VD-1000MB-FZ-520
Serial Number: 07860309
UID: 4E455441:50502020:56442D31:3030304D:422D465A:2D353230:30
373836:30333039:00000000:00000000
BPS: 520
Physical Size: 1.00GB
Position: data
Checksum Compatibility: block
Aggregate: aggr1
Plex: plex0

Paths:
LUN Initiator Side Target Side
Link
Controller Initiator ID Switch Port Switch Port
Acc Use Target Port TPGN Speed I/O KB/s IOPS
```

Zeroing the disks.

```
prodcluster::> storage disk zerospare -owner prodcluster-01
```

Fail the disks.

```
prodcluster::> storage disk show -broken
There are no entries matching your query.

prodcluster::>
prodcluster::>
prodcluster::>
prodcluster::>
prodcluster::> storage disk fail -disk NET-1.30 -immediate

Warning: The system will not prefail the disk and its contents will not be
copied to a replacement disk before being failed out. Do you want to
fail out the disk immediately? {y|n}: y

prodcluster::>
```

```
10.1.1.130 - PuTTY
prodcluster::> storage aggregate show-status -aggregate aggr1

Owner Node: prodcluster-01
Aggregate: aggr1 (online, raid4, reconstruct) (block checksums)
Plex: /aggr1/plex0 (online, normal, active, pool0)
RAID Group /aggr1/plex0/rg0 (reconstruction 10% completed, block checksums)

Position Disk Pool Type RPM Usable Physical
atus -----
-----
parity NET-1.8 0 FCAL 15000 1020MB 1.00GB (n
ormal)
data NET-1.14 0 FCAL 15000 1020MB 1.00GB (r
econstruction 10% completed)
data NET-1.9 0 FCAL 15000 1020MB 1.00GB (n
ormal)
data NET-1.15 0 FCAL 15000 1020MB 1.00GB (n
ormal)
data NET-1.10 0 FCAL 15000 1020MB 1.00GB (n
ormal)
5 entries were displayed.

prodcluster::>
```

prodcluster - NetApp OnCommand System Manager

Dashboard Volumes SVMs Network Hardware and Diagnostics Protection Configurations

### Aggregates

| Name      | Node           | Type     | Used (%) | Available Space | Used Space | Total Space | Volume Count | Disk Count | Flash Pool |
|-----------|----------------|----------|----------|-----------------|------------|-------------|--------------|------------|------------|
| aggr0     | prodcluster-01 | Standard | 54       | 1.54 GB         | 1.8 GB     | 3.34 GB     | 1            | 6          | -NA-       |
| aggr1     | prodcluster-01 | Standard | 1        | 3.5 GB          | 20.46 MB   | 3.52 GB     | 1            | 5          | -NA-       |
| depi2aggr | prodcluster-01 | Standard | 1        | 3.5 GB          | 20.45 MB   | 3.52 GB     | 1            | 5          | -NA-       |

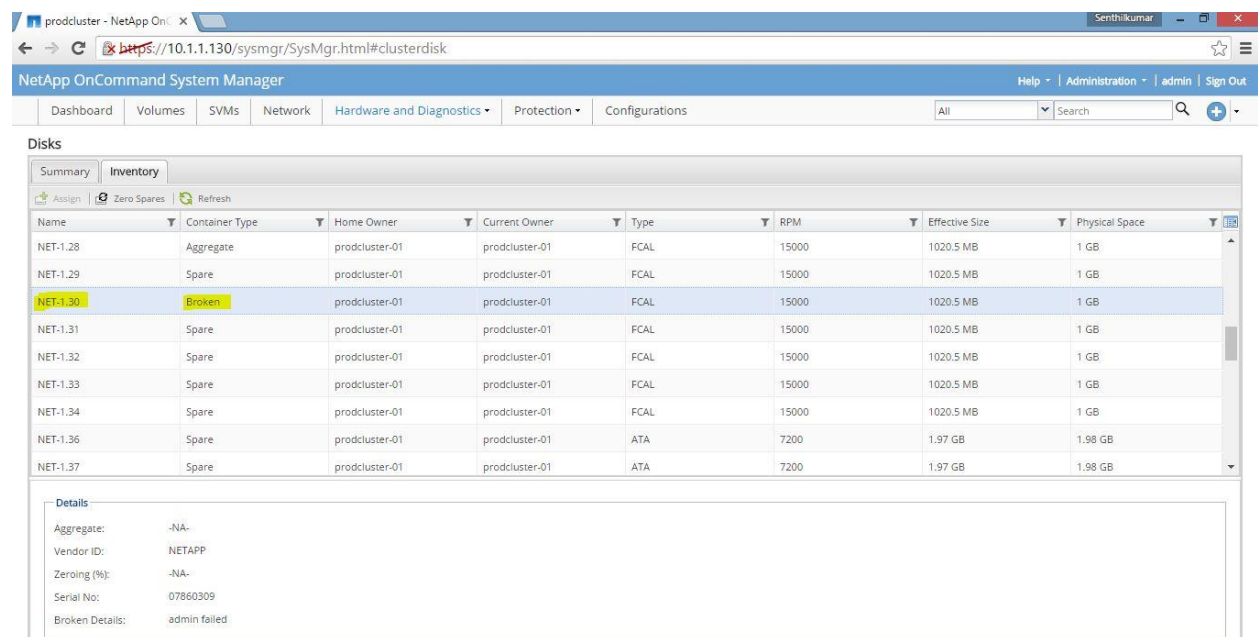
| Name     | Disk Type | Physical Size | Usable Size | Disk Status        | Plex Name | Plex Status | Raid Group | Raid type | Storage Pool |
|----------|-----------|---------------|-------------|--------------------|-----------|-------------|------------|-----------|--------------|
| NET-1.8  | FCAL      | 1 GB          | 1020.5 MB   | Normal             | plex0     | Online      | rg0        | RAID4     | -NA-         |
| NET-1.14 | FCAL      | 1 GB          | 1020.5 MB   | Reconstruction 14% | plex0     | Online      | rg0        | RAID4     | -NA-         |
| NET-1.9  | FCAL      | 1 GB          | 1020.5 MB   | Normal             | plex0     | Online      | rg0        | RAID4     | -NA-         |
| NET-1.15 | FCAL      | 1 GB          | 1020.5 MB   | Normal             | plex0     | Online      | rg0        | RAID4     | -NA-         |

## Listing the Broken disks.

```
prodcluster::> storage disk show -broken
Original Owner: prodcluster-01
Checksum Compatibility: block

Physical                                     Usable
Disk                                     Size
Size
-----
NET-1.30                                admin failed v0 - 26 A FAILED FCAL 15000 1020MB
1.00GB

prodcluster::>
prodcluster::>
prodcluster::>
```



The screenshot shows the NetApp OnCommand System Manager web interface. The browser address bar displays `https://10.1.1.130/sysmgr/SysMgr.html#clusterdisk`. The navigation bar includes links for Dashboard, Volumes, SVMs, Network, Hardware and Diagnostics, Protection, and Configurations. The 'Disks' section is active, showing a table of disk information. The table has columns for Name, Container Type, Home Owner, Current Owner, Type, RPM, Effective Size, and Physical Space. The disk 'NET-1.30' is highlighted in blue and marked as 'Broken'. Below the table, the 'Details' section shows information for the selected disk, including Aggregate, Vendor ID, Zeroing (%), Serial No, and Broken Details.

| Name     | Container Type | Home Owner     | Current Owner  | Type | RPM   | Effective Size | Physical Space |
|----------|----------------|----------------|----------------|------|-------|----------------|----------------|
| NET-1.28 | Aggregate      | prodcluster-01 | prodcluster-01 | FCAL | 15000 | 1020.5 MB      | 1 GB           |
| NET-1.29 | Spare          | prodcluster-01 | prodcluster-01 | FCAL | 15000 | 1020.5 MB      | 1 GB           |
| NET-1.30 | Broken         | prodcluster-01 | prodcluster-01 | FCAL | 15000 | 1020.5 MB      | 1 GB           |
| NET-1.31 | Spare          | prodcluster-01 | prodcluster-01 | FCAL | 15000 | 1020.5 MB      | 1 GB           |
| NET-1.32 | Spare          | prodcluster-01 | prodcluster-01 | FCAL | 15000 | 1020.5 MB      | 1 GB           |
| NET-1.33 | Spare          | prodcluster-01 | prodcluster-01 | FCAL | 15000 | 1020.5 MB      | 1 GB           |
| NET-1.34 | Spare          | prodcluster-01 | prodcluster-01 | FCAL | 15000 | 1020.5 MB      | 1 GB           |
| NET-1.36 | Spare          | prodcluster-01 | prodcluster-01 | ATA  | 7200  | 1.97 GB        | 1.98 GB        |
| NET-1.37 | Spare          | prodcluster-01 | prodcluster-01 | ATA  | 7200  | 1.97 GB        | 1.98 GB        |

| Details         |              |
|-----------------|--------------|
| Aggregate:      | -NA-         |
| Vendor ID:      | NETAPP       |
| Zeroing (%):    | -NA-         |
| Serial No:      | 07860309     |
| Broken Details: | admin failed |

Unfail the disks.

```
10.1.1.130 - PuTTY
prodcluster:.*> storage disk show -broken
Original Owner: prodcluster-01
Checksum Compatibility: block

Physical                                     Usable
Disk                                     Size
Size
-----
NET-1.30                                1020MB
1.00GB      admin failed v0      - 26      A FAILED FCAL 15000

prodcluster:.*>
prodcluster:.*>
prodcluster:.*> storage disk unfail -disk NET-1.30

Warning: Failed disk "NET-1.30" may have aggregate labels and file system data
present. In that case, this command will attempt to bring this disk
back into the aggregate with which this disk had formerly been
associated and preserve file system data. Are you sure you want to
continue with disk unfail? {y|n}: y

prodcluster:.*> storage disk show -broken
There are no entries matching your query.
```